



Nursing Study Guide: Fluid, Electrolytes & Perioperative Care Cred 101

Geometry (Palm Beach Preparatory Charter Academy)



messages.pdf_cover_qr_code_label

Fluid and Electrolytes

- **Signs of Fluid Volume Deficit:** Dry mucous membranes, poor skin turgor, oliguria, tachycardia, hypotension.
- **Hypervolemia Signs:** Bounding pulse, edema, JVD, crackles in lungs.
- **Electrolyte Basics:**
 - **Potassium:** Primarily intracellular; hyperkalemia may cause arrhythmias.
 - **Magnesium:** Linked to tremors, confusion if low; found in intracellular fluid.
 - **Sodium/Chloride:** Extracellular; check sodium for neuro changes.
 - **Hypocalcemia Signs:** Muscle twitching, tetany, Chvostek's sign.
- **Key Terms:**
 - **Osmotic Pressure:** Increased in hyperglycemia (DKA).
 - **Oncotic Pressure:** Related to fluid in the abdominal cavity (ascites).
 - **Metabolic Acidosis:** Potassium shifts out of cells, increasing hyperkalemia risk.

2. Catheterization and CAUTI Prevention

- **CAUTI Prevention:** Remove Foley ASAP, use aseptic technique.
- **Determining Foley Size:** Measured in French.
- **Measuring Intake & Output:** Essential for monitoring fluid balance.

3. Gastrointestinal Disorders

- **GERD Management:** Small frequent meals, avoid hot/spicy foods, no eating 2-3 hours before bed, PPIs, prokinetic agents, elevate head of bed, avoid fatty foods, caffeine, chocolate, alcohol.
- **Crohn's Disease:** Assess stool and dehydration; corticosteroids for inflammation.
- **Pancreatitis:**
 - **Symptoms:** Severe LUQ or epigastric pain radiating to back; knee-chest position relieves pain.
 - **Treatment:** Opioids (morphine, fentanyl), anticholinergics, gastric acid reducers, pancreatic enzymes for chronic cases.
- **Diverticulitis:** Low-residue diet, avoid seeds.

4. Reproductive Health

- **Myasthenia Gravis:** Persistent fatigue, muscle weakness affecting eyes, face, swallowing.
- **Prostate Gland Function:** Provides nutrients to sperm.
- **BPH:** Evaluate symptoms and monitor.

5. Pregnancy and Neonatal Care

- **GTPAL:** Gravida (G), Term (T), Preterm (P), Abortions (A), Living children (L).
- **Placenta Previa:** Placenta close to cervix.
- **HELLP Syndrome:** Thrombocytopenia, elevated liver enzymes, low platelets.

- **Hyperemesis Gravidarum:** Severe vomiting, dehydration, electrolyte imbalance.
- **Postpartum Hemorrhage:** Assess and treat using uterine massage, methylergonovine (Methergine).
- **Involution:** Organs returning to pre-pregnancy state.
- **Effacement:** Cervical thinning, measured in %.
- **Cerclage:** Prevents preterm labor by supporting an incompetent cervix.
- **Hydatidiform Mole:** No fetal heart activity; monitor HcG levels.

6. Mental Health and Psych Disorders

- **Therapeutic Communication:** Open-ended questions, validate feelings, ask about self-harm.
- **Depression:** Screen using DSM-5; persistent sadness or low mood.
- **Suicidal Ideation:** Use suicide precautions, notify HCP.
- **Generalized Anxiety Disorder:** Persistent restlessness, fatigue.
- **Social Phobia:** Fear of social situations, avoidance due to fear of judgment.
- **Eating Disorders:**
 - **Anorexia:** Fear of weight gain, body dysphoria, malnutrition.
 - **Bulimia:** Binge-eating with compensatory behaviors (e.g., purging).
- **Psych Medications:**
 - **SSRI Toxicity:** Never stop abruptly, monitor for suicidal thoughts, avoid St. John's Wort.
 - **Serotonin Syndrome:** High fever, hyperreflexia, tachycardia.
 - **MAOIs:** Avoid tyramine-rich foods (aged cheese, fermented foods).
 - **Lamotrigine:** Treats bipolar; watch for Stevens-Johnson syndrome.
- **Schizophrenia:** Hallucinations, delusions, word salad, flat affect.
- **Baker Act:** Law allowing involuntary examination of individuals in mental health crises.

7. Endocrine and Metabolic Disorders

- **Diabetes:**
 - **3 P's:** Polyuria, polydipsia, polyphagia.
 - **Type 1 Management:** Lifelong insulin therapy.
- **SIADH:** Treat with 3% sodium chloride for hyponatremia.
- **Hypoglycemia:** Signs in infants include hypoactivity and excessive crying.

8. Labor and Delivery

- **Stages of Labor:** Monitor fetal heart rate and check for accelerations.
- **Assessing Fetal Position:** Vertex indicates head-down position.
- **Intrapartum:** After water breaks, check fetal heart rate to detect possible cord compression.
- **Postpartum:**
 - **Fundal Massage:** For boggy uterus.
 - **Fundal Height:** Correlates to weeks of gestation (20 weeks at the umbilicus).
 - **Lochia Stages:** 6 days post-op, expect serosa discharge.

9. Newborn and Infant Care

- **Vaccines:** First vaccine is typically Hep B.
- **PKU:** Avoid milk, eggs, dairy, and meat.
- **Safe Sleep:** Supine position, monitor for fever >100.4°F, jaundice.
- **Circumcision Care:** Apply Vaseline gauze to prevent sticking.
- **Congenital Heart Defects:** Pulse ox screening.

Newborn and Pediatric Study Guide

1. Newborn Assessment and Reflexes

- **Temperature & Patency Check:** Use rectal temp to check rectum patency; observe for imperforate anus.
- **Meconium:** Should pass within 24 hours.
- **APGAR Score:** Assess at 1 and 5 minutes; normal score around 9.
- **Common Findings:**
 - **Mongolian Spot:** Bluish spot on back.
 - **Vernix:** White substance on skin at birth.
 - **Moro Reflex:** Startle reflex—arms and legs extend, then draw back with clenched fists.
 - **Babinski Reflex:** Toes hyperextend or dorsiflex when foot is stroked.
- **Head Findings:**
 - **Caput Succedaneum:** Soft swelling on scalp, crosses suture lines.
 - **Cephalhematoma:** Blood collection between skull bones, does not cross suture lines.
- **Skin Color:**
 - **Acrocyanosis:** Blue hands and feet, common in newborns.
- **Barlow Maneuver:** Checks for hip dysplasia; assess hip stability and gluteal folds.
- **Feeding:** Feed every 2-3 hours.

2. Cold Stress & Respiratory Care

- **Cold Stress Effects:** Hypoglycemia, respiratory distress.
- **Respiratory Distress Signs:** Tachypnea, grunting, nasal flaring, retractions.
- **Interventions for Tetralogy of Fallot:** If cyanosis is noted, place the infant in knee-chest position.
- **Surfactant:** Fatty acid substance aiding lung function.

3. Congenital Conditions and Pediatric Syndromes

- **Tetralogy of Fallot:** Four defects (overriding aorta, right ventricular hypertrophy, VSD, pulmonary stenosis); signs include cyanosis, difficulty feeding.
- **Spina Bifida Cystica:** Two types affecting the spinal cord.
- **Omphalocele:** Abdominal wall defect.

- **Trisomy Conditions:**
 - **Trisomy 19 (Edwards Syndrome):** Severe developmental issues.
 - **Trisomy 21 (Down Syndrome):** Single palmar crease, extra neck skin, slanted eyes.
 - **Turner's Syndrome:** Affects only females; distinctive physical features.
- **PKU:** Missing enzyme for amino acid breakdown; requires testing within 72 hours of birth, managed with diet free from certain proteins.
- **Cystic Fibrosis:** Thick lung mucus; treatment includes CPT, enzymes on food, high-calorie and high-protein diet.

4. Nutritional & Developmental Care

- **Vitamin K Injection:** Given to help blood clotting.
- **Special Diets:**
 - **Celiac Disease:** Avoid gluten (wheat, rye, barley).
 - **PKU:** Diet excludes foods with high protein.
- **Feeding for Cleft Lip/Palate:** Use special nipple, feed at an angle, keep upright post-feeding.

5. Conditions and Interventions

- **Congenital Heart Defect Signs:** Occiput prominence, small jaw/mouth.
- **Sickle Cell Disease:** Prioritize pain management, oxygenation, and IV fluids.
- **Leukemia:** Low platelet levels; bleeding precautions.
- **Muscular Dystrophy (Early Signs):** Clumsiness, frequent falls.
- **Wilms Tumor:** Kidney tumor; do not palpate abdomen.
- **Pyloric Stenosis:** Projectile vomiting, RUQ olive-shaped mass; surgical treatment is pyloromyotomy.
- **Intussusception:** Distended abdomen, jelly-like stools; barium enema is both diagnostic and therapeutic. Will have white chalky stools.

Stage 1: Labor Phases

1. **Latent (Early Labor)**
 - Cervix dilation: 1-4 cm
 - Contractions: 5-30 minutes apart, lasting 30-45 seconds
 - Mom's Response: Mild discomfort, excited, interactive with family
2. **Active Phase**
 - Cervix dilation: 4-7 cm
 - Contractions: 3-5 minutes apart, lasting 45-60 seconds (strong and long)
 - Duration: 4-8 hours, pain intensifies
3. **Transition Phase**
 - Cervix dilation: 8-10 cm
 - Contractions: 2-3 minutes apart, lasting 60-90 seconds (intense)
 - Duration: 30 minutes-2 hours, very strong contractions

Stage 2: Delivery of Baby

- Position: Place mom supine, partner and nurse assist with leg support during contractions
- Monitor: Continuous Fetal Heart Rate (FHR) assessment
- Presentation: Check presenting part (breech presentation requires C-section)

Stage 3: Delivery of Placenta

- Duration: From delivery of baby to delivery of placenta
- Placenta Assessment:
 - **Shiny/White (Schultze)**: Baby side of placenta
 - **Dirty (Duncan)**: Mother's side of placenta
- **Complications**: Check for missing parts, especially on Duncan side; if parts are missing, manual removal may be required due to hemorrhage risk

Stage 4: First Hour Postpartum

- Hemorrhage Risk: Monitor mom for postpartum hemorrhage
- Vital Signs: Every 5 minutes initially, then every 15 minutes
- Fundus Assessment:
 - Should be firm and midline; if boggy, massage and ensure bladder is empty
- Lochia (postpartum bleeding):
 - **Lochia Rubra**: Red, for first few days
 - **Lochia Serosa**: Pink, lasts 2-3 weeks
 - **Lochia Alba**: White/yellow
- Comfort: Apply ice packs for pain relief

Assisting with Delivery of Baby

1. **Preparation:**
 - Set up the delivery room and instruments, covered with a sterile drape
 - Break down bed for mom's positioning to push
 - Prepare meds: Pitocin, 2 bags of LR, Narcan
 - Check resuscitation equipment (O2, suction, ambu bag for mom and baby)
 - Turn on warmer, place towels and baby hat in the warmer
2. **Immediate Care After Baby's Birth:**
 - Suction: Suction mouth, then nose to clear fluids
 - Newborn Care:
 - Dry baby thoroughly, use bulb syringe as needed, and place baby with mom if stable
 - VS & APGAR at 1 and 5 minutes
 - Weight, measurements, and blood sugar check if mom is diabetic
 - Identification: Place armbands on baby, do footprints
 - Placenta Inspection: Save cord blood for lab tests if needed

BUBBLE HEE Framework

1. **B: Breasts**

- Assess for:
 - **Consistency:** Soft (filling) or firm (engorgement)
 - **Tenderness:** Tender or nontender
 - **Temperature:** Warm or cool
 - **Nipples:** Check for cracking or breakdown

2. **U: Uterus (Fundus)**

- Immediately after delivery, the fundus should be:
 - Firm and midline
 - Measured in relation to the umbilicus (e.g., U +1, U -1)
 - 6-12 hours postpartum: Fundus at the level of the umbilicus
 - Postpartum day 2: Should be U -1
- Massage technique: Upper hand on top of fundus, lower hand supporting uterus above the symphysis pubis

3. **B: Bladder**

- Assess for distention:
 - Should void within 6 hours after delivery; may require straight catheterization if unable to void
 - If boggy and deviated, encourage bladder emptying

4. **B: Bowel**

- Bowel sounds may be hypoactive initially:
 - Expected bowel movement by day 2 or day 3
 - May need stool softeners; encourage increased fiber and fluids

5. **L: Lochia**

- Assess type and amount:
 - **Type:**
 - Rubra (red): 3-4 days
 - Serosa (pink): 4-10 days
 - Alba (white): 10+ days
 - **Amount:**
 - Scant: 1 inch stain in 1 hour
 - Light: <4 inch stain
 - Moderate: 6 inches
 - Heavy: Saturated
- Note earthy odor; no clots should be present

6. **E: Episiotomy**

- Assess for REEDA (Redness, Edema, Ecchymosis, Drainage, Approximation):
 - Treatment: Ice for the first 24 hours, peri bottle with each void or bowel movement, sitz baths
- For C-section: Assess REEDA and perform gentle fundal checks

7. **H: Hemorrhoids**

- Assess for size and swelling:
 - Treatment: Ice, Tucks, lidocaine spray

8. **E: Extremities**

- Assess for edema, pain, and pulses:
 - Palpate calves and behind the knees; check Homan's sign for DVT

9. **E: Emotions**

- Common to experience postpartum blues for 3-5 days due to hormonal fluctuations:
 - Monitor for signs of postpartum depression, such as lack of eye contact or interaction with the baby

Term

- **Involution:** The process of reproductive organs returning to their prepregnant state.

Head-to-Toe Newborn Assessment

1. **Initial Assessment**

- **Vital Signs:** Check respirations and heart rate for 1 minute first.
- **Overall Appearance:** Observe color, symmetry, and respiratory effort.
- **Skin Assessment:** Look for rashes, birthmarks, hemangiomas, milia, and check buttocks for Mongolian spots.

2. **Head**

- **Fontanel:** Posterior and anterior fontanel should be flat and soft.
 - **Bulging:** May indicate increased ICP (intracranial pressure).
 - **Sunken:** May indicate dehydration.
- **Cephalohematoma:** Does not cross suture line; bleeding between skull and periosteum, resolving in weeks to months.
- **Caput Succedaneum:** Crosses suture line; edema resolves in a few days.

3. **Eyes**

- Assess for symmetry and color.

4. **Nose**

- Check for patency and nasal flaring.

5. **Mouth**

- Inspect for cleft lip and palpate for cleft palate.
- Assess sucking reflex.

6. **Ears**

- Check for symmetry; should be in line with the eyes.

7. **Neck**

- Assess range of motion (ROM).

8. **Clavicles**

- Check for fractures.

9. Chest

- **Heart Rate and Lung Sounds:** Assess over 1 full minute.
- **Lungs:** Respiratory rate 30-60; may hear a few crackles (amniotic fluid).
- **Heart:** Rate 120-160; listen for rate, rhythm, and murmurs.

10. Circulation

- Assess brachial and femoral pulses.
- Measure blood pressure in upper and lower extremities.
- Check capillary refill over the sternum.

11. Abdomen

- Inspect, auscultate, and palpate.
- **Cord:** 2 arteries, 1 vein.

12. Genitourinary

- **Female:** Check for ambiguous genitalia; labia majora may be swollen with white discharge.
- **Male:** Check penis for hypospadias; urinary meatus should be midline.
 - Scrotum should be pink and may be swollen; testes should be descended.

13. Anus

- Check for patency; take rectal temperature.
- **Meconium:** Should be passed within the first 24 hours.

14. Hips

- Assess range of motion (ROM).
- **Barlow Maneuver:** Adduct hips and apply posterior pressure; a "clunk" indicates dislocated femoral head.
- **Ortolani Maneuver:** Abduct leg and rotate out to relocate dislocated hip; a "clunk" indicates hip relocation.

15. Musculoskeletal

- Assess tone and ROM of extremities.
- Count fingers and toes.

16. Back

- Check spine; look for any dimpling in the skin.
- Assess gluteal folds for symmetry (uneven folds may indicate hip problems).

Breastfeeding Positions

1. Laid-Back Nursing Position

- **Description:** Relaxed position where the mother reclines slightly, allowing the baby to find the breast naturally.
- **Benefits:** Encourages a good latch; ideal for any mother and baby.

2. Cradle Hold

- **Description:** The baby is cradled in the mother's arms, resting along the forearm.
- **Benefits:** Comfortable once the baby has latched on; suitable for many mothers.

3. Cross-Cradle Hold

- **Description:** The mother uses the opposite arm to support the baby, allowing better control of the baby's head.
- **Benefits:** Particularly helpful for premature infants, newborns, or babies who have difficulty latching.

4. Football Hold

- **Description:** The baby is held under the mother's arm, resembling holding a football.
- **Benefits:** Great for nursing twins, beneficial for mothers recovering from a C-section, or those with large breasts or flat/inverted nipples.

5. Side-Lying Position

- **Description:** The mother lies on her side, with the baby positioned facing her.
- **Benefits:** Excellent for nighttime feedings and recovery after a C-section.

Breastfeeding Guide

Colostrum:

- **Definition:** The first pre-milk produced by the mother.
- **Importance:** Rich in antibodies and immunoglobulins that strengthen the baby's immune system.

Milk Transition:

1. **Transitional Milk:**
 - Begins 2-3 days after birth, lasting about 2 weeks.
2. **Mature Milk:**
 - Produced after the transitional phase.

Breastfeeding Timing:

- **Optimal Start:** Within the first hour of life when the baby is more alert.

Positioning:

- **Skin-to-Skin:** Position the mother skin-to-skin with the baby.
- **Latching:** Stroke the baby's cheek with the nipple and ensure the whole areola is in the baby's mouth.

Feeding Schedule:

- Nurse every 2-3 hours around the clock.
- Duration: 10-15 minutes on one side before switching.
- Next feeding should start on the side finished during the previous session.

Breaking Suction:

- Teach the mother to break suction using her finger and to burp the baby after each breast.

Indicators of Adequate Intake:

- 6-8 wet diapers per day.
- Presence of let-down reflex.
- Audible swallowing during feeding.
- Steady weight gain.
- Baby appears content after feedings.

Jaundice in Newborns

Definition:

Jaundice is a yellowish discoloration of the skin and eyes in newborns caused by elevated levels of bilirubin.

Bilirubin:

- **Description:** A yellowish pigment produced during the breakdown of red blood cells.
- **Function:** Normally processed by the liver and excreted from the body.

Causes:

- In utero, the mother removes bilirubin for the fetus. After birth, the newborn's liver may not be mature enough to handle bilirubin elimination, leading to jaundice.

Progression:

- **Mild Jaundice:** Seen on the face and chest.
- **Severe Jaundice:** Progresses to arms, trunk, and legs.

Risk Factors:

- Premature birth
- Feeding difficulties
- Ethnic backgrounds (East Asian and Mediterranean descent)

- Presence of bruising
- History of jaundice in siblings
- Darker skin tones (as jaundice is harder to detect, check gums and lips)

Complications:

If untreated, severe jaundice can lead to:

- Kernicterus (brain damage)
- Cerebral palsy
- Hearing loss

Treatment:

1. **Phototherapy:**
 - Undress the baby, leaving only the diaper on.
 - Place the baby under UV phototherapy with eye protection.
2. **Increase Milk Intake:**
 - Encourages bilirubin elimination.
3. **Blood Exchange Transfusion:**
 - Considered for severe cases.

Monitoring:

- Bilirubin levels peak at 3-5 days old, making follow-up with a pediatrician crucial after discharge to ensure proper management.

Umbilical Cord Care Guide

Care Instructions:

- **Diaper Placement:** Fold the diaper under the umbilical cord to avoid covering it.
- **Cleanliness:** Keep the area clean and dry.
- **Bathing:** Do not submerge the baby in water; give sponge baths until the cord stump falls off.

Fall Off Timeline:

- The umbilical cord stump typically falls off within **10 days**.

Signs of a Healthy Umbilical Cord Stump:

- The stump should appear dry and may have a small amount of yellowish discharge.
- Avoid any signs of redness, swelling, or foul odor, which may indicate infection.

Pulse Oximeter Application in Newborns

Purpose:

- To check oxygenation levels and help rule out critical congenital heart defects (CCHD).

Placement:

1. **Right Hand:** Place one pulse oximeter probe on the pinky side of the right hand.
2. **Foot:** Place another probe on either foot, also on the pinky side.

Application Tips:

- Ensure the light emitter on the probe aligns with the photodetector.
- Use a new probe for each patient to prevent cross-contamination.
- Make sure there are no gaps between the sensor and the skin for accurate readings.
- Keep the newborn warm, as movement and shivering can affect pulse oximetry readings.

Monitoring:

- Look for the pleth waveform, which should match the heart rate (check apical pulse).

Applying a Urine Bag

Preparation:

1. Clean the area using three towelettes or soap and water, then rinse.

For Females:

- Clean the labia and meatus from front to back.

For Males:

- Clean the tip of the penis to the scrotum three times (do not forcefully retract the foreskin).

Placement:

- **Females:** Position the bag on the labia, adhering the creased part to the perineal area.
- **Males:** Place the penis and scrotum in the bag, adhering the creased part to the perineal area.

Securing the Bag:

- Place a diaper over the bag to prevent it from being pulled off.

Collecting Specimen:

- If the bag has a tab, rip it to pour urine into a specimen cup. If not, seal the bag and place it in a specimen cup.
-

Circumcision Care

Gomco Method:

- Apply petroleum gauze to the head of the penis to prevent sticking to the diaper.

Plastibell Method:

- Clean the penis with water after each void and bowel movement (squeeze water over the penis). The plastic ring prevents the penis from sticking to the diaper.

Diaper Placement:

- Apply a clean diaper snugly, folding it under the cord.

Post-Operative Care:

- Avoid placing the baby on their stomach for 3-5 days.
- Monitor for signs of infection, bleeding, increased temperature, and urinary retention.

Infant Identification Upon Delivery

Bracelet Information:

Ensure that each bracelet is filled out with the following details:

1. **Mother's Name:** Clearly print the full name of the mother.
2. **Sex of Infant:** Indicate whether the infant is male or female.
3. **Doctor's Name:** Include the name of the attending physician.
4. **Date of Birth (DOB):** Write the infant's date of birth.
5. **Time of Birth:** Note the exact time the infant was born.
6. **Medical Record Number (MR #):** Include the hospital's medical record number for the infant

Newborn Footprint Procedure

Timing:

- Complete the footprint 1-2 hours after birth.

Materials Needed:

- Ink pad
- Paper form for footprints

Procedure:

1. **Hold the Foot:** Gently grasp the newborn's foot by the ankle.
2. **Ink the Foot:** Press the foot onto the ink pad, ensuring all surfaces are well covered with ink.
3. **Make the Print:** Firmly press the inked foot onto the designated area of the paper form in a heel-to-toe motion.
4. **Fingerprinting:** Simultaneously, take the mother's fingerprints for identification.

Newborn Screening Specimen Collection and Important Conditions

Newborn Screening Overview

- **Purpose:** A blood test performed before discharge to check for rare genetic, hormone-related, and metabolic conditions that could lead to serious health issues.

Key Conditions Screened:

1. Phenylketonuria (PKU)

- **Definition:** A metabolic disorder caused by a defect in the enzyme responsible for breaking down phenylalanine, an amino acid.
- **Signs/Symptoms:**
 - Developmental delays and permanent intellectual disability if untreated.
 - Fair skin, light hair, and blue eyes due to melanin disruption.
 - May present with seizures, growth problems, behavioral issues, skin rashes, and a musty odor to breath, skin, or urine.
- **Treatment:** Strict lifelong diet low in phenylalanine. Babies require phenyl-free formula, and older children and adults should avoid high-protein foods and aspartame.

2. Galactosemia

- **Definition:** A rare, hereditary disorder that impairs the body's ability to convert galactose (found in milk) into glucose due to a deficiency of the enzyme galactose-1-phosphate uridylyl transferase (GALT).
- **Signs/Symptoms:**
 - Normal appearance at birth followed by poor feeding, vomiting, and jaundice within days or weeks.
 - May lead to hepatomegaly, growth failure, ascites, and diarrhea.

- Risk of cataracts, liver failure, and intellectual disability if untreated.
- **Treatment:** Early diagnosis and a strict lactose-restricted (dairy-free) diet are essential to prevent severe complications.

3. **Maple Syrup Urine Disease (MSUD)**

- **Definition:** A life-threatening metabolic disorder characterized by the inability to break down branched-chain amino acids (leucine, isoleucine, valine).
- **Signs/Symptoms:**
 - Urine, sweat, or earwax that smells like maple syrup.
 - Poor feeding, vomiting, irritability, and lethargy within the first few days of life.
 - Can progress to seizures, convulsions, and coma without treatment.
- **Treatment:** A diet very low in protein and BCAAs; regular medical management is crucial to prevent complications.

4. **Congenital Hypothyroidism**

- **Definition:** A severe deficiency of thyroid hormone present at birth, leading to impaired neurological function and growth.
- **Signs/Symptoms:**
 - Lack of weight gain, fatigue, lethargy, and poor feeding.
 - Thickened facial features, abnormal bone growth, excessive sleep, and a large tongue.
 - Jaundice and low muscle tone may also be present.
- **Causes:** Can result from a missing or underdeveloped thyroid gland, genetic defects, or maternal iodine deficiency during pregnancy.
- **Treatment:** Routine screening for T4 and TSH levels. Treated with levothyroxine; early treatment within the first four weeks is vital to prevent permanent intellectual disability.

Heel Stick Procedure for Blood Sample Collection

1. **Preparation:**

- Wash hands and don gloves.
- Confirm patient identity.
- Position the limb lower than the heart to enhance blood flow.

2. **Warm the Heel:**

- Apply heel warmers for 3 minutes to promote blood circulation.

3. **Clean the Heel:**

- Use an alcohol swab to cleanse the heel and allow it to dry.

4. **Perform the Heel Stick:**

- Make the puncture to collect blood for screening.

VEAL CHOP MINE

- **Variable Decelerations**
 - **Cause:** Cord compression
 - **Management:** Maternal repositioning
- **Early Decelerations**
 - **Cause:** Head compression
 - **Management:** Identify labor progress (no interventions needed)
- **Accelerations**
 - **Cause:** Okay! (Indicates fetal well-being)
 - **Management:** No interventions needed
- **Late Decelerations**
 - **Cause:** Placental insufficiency
 - **Management:** Execute interventions (e.g., maternal repositioning, oxygen, IV fluids)

Final Review Part 1

- **Suicidal Ideation**
 - **Understanding:** Know reasons for ideation; don't hesitate to ask patients directly about their feelings.
- **Hallucinations**
 - **Context:** Common in psychiatric patients, especially during detox.
- **Reporting Secrets**
 - **Policy:** If a patient discloses something like drug possession, inform them you must report it to your supervisor.
- **Serotonin Syndrome**
 - **Definition:** Toxicity from overuse of SSRIs.
 - **Symptoms:** Dry mouth, sweating, fever, tremors, muscle rigidity.
- **Haloperidol**
 - **Use:** Antipsychotic for schizophrenia and "sundowning."
- **Anorexia and Bulimia**
 - **Anorexia:** Fear of gaining weight; assess menstruation for amenorrhea.
 - **Bulimia:** Don't allow bathroom access alone post-meal.
- **Substance Abuse Detox**
 - **Signs of Withdrawal:** Disorientation, tachycardia, tremors, sweating.

- **Management:** Use CWA protocol; administer benzodiazepines.
- **Schizophrenia Symptoms**
 - **Signs:** Hallucinations, delusions, "word salad," flat affect.
- **Math and ABGs**
 - **Skills:** Be proficient in calculations and interpreting arterial blood gases.
- **OB and FHR Strips**
 - **Indicators:** Understand fetal heart rate patterns and what they indicate.
- **Magnesium Toxicity**
 - **Symptoms:** Absence of reflexes, hypertension, headache, blurred vision, pedal edema. Monitor in pregnant patients.
- **Newborn Vaccinations**
 - **First Vaccine:** Hepatitis B.
- **Prenatal Vitamins**
 - **Essential Component:** Folic acid to prevent neural tube defects.
- **Intussusception**
 - **Definition:** Bowel collapsing on itself; can be resolved with a barium enema.
 - **Symptoms:** Currant jelly stools; assess for hydration.
- **Asthma in Children**
 - **Action:** Check oxygen saturation first in a 7-year-old with trouble breathing after playing.
- **Down Syndrome Characteristics**
 - **Traits:** Slanting eyes, single palmar crease, flat facial features, short neck.
- **Cystic Fibrosis Effects**
 - **Impact:** Affects lungs with sticky mucus; administer pancreatic enzymes with food.
- **Hepatitis A**
 - **Transmission:** Spread through stool.
- **Peptic Ulcer Management**
 - **Treatment:** PPIs; diagnosis via endoscopy.
- **Respiratory Disturbances**
 - **Understand:** Differences between respiratory alkalosis and acidosis, their causes, and effects.

- **HIPAA Compliance**
 - **Scenario:** If a parent asks about their child's status, remember confidentiality laws.
- **Myasthenia Gravis**
 - **Symptoms:** Persistent fatigue and weakness.
- **Seizure Medication for Infants**
 - **Medication:** Keppra.
- **Epilepsy Management**
 - **Guidelines:** Patients must take medication for life or as directed by a doctor.

Urinary Catheterization Study Guide

Definition:

- A hollow tube placed through the urethra into the bladder.
-

Purposes of Urinary Catheterization:

1. Monitor strict intake and output (I&O).
 2. Relieve urinary obstruction.
 3. Provide pre-operative and post-operative care.
 4. Aid patients with neurological conditions preventing complete bladder emptying.
 5. Assess post-void residual when a bladder scanner is unavailable.
-

Types of Catheters:

1. **Straight Catheter:** Single lumen for short-term use.
 2. **Double Lumen Catheter:** Indwelling retention catheter for long-term use.
 3. **Triple Lumen Catheter:** For continuous closed irrigation.
-

French Measurement:

- Foley catheters are measured in French units (1 French = 0.33 mm).
 - An 18 French catheter is approximately 6 mm in diameter.

- **Size Guidelines:**
 1. **Children:** 8-10 FR.
 2. **Females and Males:** 14-16 FR.
 3. **Males with Enlarged Prostate:** May need a smaller size or a coude catheter (slightly bent tip) to bypass obstructions.
 4. **Post-Urological Surgery:** Larger sizes may be used to facilitate hematuria and clot passage.
-

Suprapubic Catheters:

- Indicated for patients with:
 - Enlarged prostates.
 - Urethral strictures.
 - Post-urologic surgery.
 - **Care:** Clean around the catheter with aseptic technique in a circular motion (cleanest to dirtiest).
-

Pericare Importance:

- Essential to prevent urinary tract infections (UTIs).
 - Mandatory prior to catheter insertion.
-

CAUTI (Catheter-Associated Urinary Tract Infection)

Prevention Strategies:

1. Use catheters only when absolutely necessary.
2. Insert catheters using sterile technique.
3. Maintain urinary catheters:
 - Perform perineal care regularly.
 - Ensure no kinks in the tubing.
 - Keep the drainage bag below the bladder.
 - Monitor for signs and symptoms of UTI.
4. Remove catheters as soon as possible.

Psychiatric Nursing Study Guide

Antidepressants

Rules for Antidepressants:

1. **Suicide Risk:** Increased risk during the first few weeks, especially in young adults (18-24) and children.
 - **Nursing Interventions:**
 - Clarify new prescriptions and monitor for suicidal thoughts.
 - Observe for unusual behavior and worsening depression.
 - Watch for sudden mood changes (e.g., increased energy with improved mood).
2. **Onset:** Slow onset (2-4 weeks); do not stop after 1 week. Taper off medications.
3. **Drug Interactions:**
 - Avoid mixing **SSRIs** with St. John's wort.
 - Avoid **MAOIs** with other antidepressants; a 2-week washout period is necessary.

SSRI (Selective Serotonin Reuptake Inhibitors):

- **Mechanism:** Inhibit serotonin reuptake, increasing serotonin levels.
- **Side Effects:** Insomnia, weight gain, sexual dysfunction, seizures.
- **Serotonin Syndrome:** Risk increased when mixed with certain drugs (St. John's wort, MAOIs, tramadol).
 - **Symptoms:** Sweating, rigidity, increased heart rate.

SNRI (Serotonin-Norepinephrine Reuptake Inhibitors):

- **Mechanism:** Similar to SSRIs but also block norepinephrine effects.
- **Notable Drug:** Duloxetine (Cymbalta) - treats depression and pain from fibromyalgia.

TCA (Tricyclic Antidepressants):

- **Mechanism:** Block serotonin and norepinephrine reuptake.
- **Adverse Effects:** Dry mouth, urinary retention, cardiovascular issues.
- **TCA Toxicity Symptoms:**
 - Tardive dyskinesia, akinesia, akathisia, pseudoparkinsonism.

MAOI (Monoamine Oxidase Inhibitors):

- **Mechanism:** Increase norepinephrine, dopamine, and serotonin.
- **Notable Medications:** Phenelzine, Selegiline, Isocarboxazid.
- **Acronym MAOI:**
 - **M:** Massive hypertension crisis risk.
 - **A:** Avoid tyramine-containing foods (e.g., aged cheeses, cured meats, fermented products).
 - **O:** Avoid certain OTC medications (NSAIDs, antacids).

- **O:** No other antidepressants.
 - **I:** Increased suicide risk.
-

Mood Stabilizers

Lithium:

- **Mechanism:** Enhances serotonin and/or GABA functions.
 - **Therapeutic Range:** 0.6-1.2 mEq/L (narrow therapeutic window).
 - **Contraindications:** Avoid diuretics, fluoxetine, NSAIDs.
 - **Monitor for:** Excessive thirst/urination, vomiting, confusion, ataxia, tremors.
 - **Teaching:** Increase fluid and salt intake, monitor weight, BUN, creatinine.
-

Anxiolytics (Antianxiety Medications)

Mechanism: Depresses CNS, increasing GABA for relaxation.

1. **Benzodiazepines** (e.g., alprazolam, temazepam):
 - **Uses:** Anxiolytic, sedative, muscle relaxant.
 - **Cautions:** Highly addictive, use short-term, do not stop abruptly.
 - **Overdose Antidote:** Flumazenil.
 2. **Barbiturates** (e.g., pentobarbital, phenobarbital):
 - **Uses:** Short-term insomnia and anxiety treatment.
 - **Risks:** Increased toxicity due to longer duration in the body.
 3. **Buspirone (Buspar):**
 - **Characteristics:** Non-sedating, takes 2-4 weeks for full effect, no withdrawal symptoms.
-

Antipsychotic Medications

Mechanism: Affect dopamine receptors to reduce psychotic symptoms.

1. **Typical Antipsychotics** (e.g., haloperidol, chlorpromazine):
 - **Uses:** Treat positive symptoms of schizophrenia (hallucinations, delusions).
2. **Atypical Antipsychotics** (e.g., aripiprazole, clozapine, quetiapine, risperidone):
 - **Uses:** Treat negative symptoms (apathy, blunted affect, anhedonia, avolition).

3. Side Effects:

- **Anticholinergic Effects:** Dry mouth, urinary retention, hypotension.
- **Extrapyramidal Symptoms:** Parkinsonism, tremors, rigidity, akathisia, tardive dyskinesia.
- **Neuroleptic Malignant Syndrome:**
 - **Symptoms:** Dyspnea, tachypnea, hypertension or hypotension, altered LOC.
 - **Treatment:** Notify MD, monitor vital signs and electrolytes, cooling measures.

What is RhoGam?

- **Definition:** RhoGam is an immune globulin injection used to prevent Rh incompatibility between an Rh-negative mother and an Rh-positive baby.
 - **Source:** Made from human blood.
 - **Importance:**
 - Most people are Rh-negative.
 - If both parents are Rh-negative, the baby will be Rh-negative.
 - If both parents are Rh-positive, the baby will be Rh-positive.
 - If the mother is Rh-negative, the baby could be either Rh-positive or Rh-negative.
-

Mechanism of Action

- **Rh Incompatibility:** If an Rh-negative mother has an Rh-positive baby, her body may recognize the Rh-positive cells as foreign and produce antibodies against them.
 - **Consequences:**
 - Can lead to anemia, kidney failure, and shock in the fetus.
 - Antibody levels can be monitored; higher levels indicate increased risk of complications.
-

Administration Guidelines

- **Dosage:** Typically 300 mcg (may vary based on blood transfer).
 - **Timing:**
 - Given at **28 weeks** gestation if the mother is Rh-negative.
 - Administered within **72 hours** after delivery of an Rh-positive baby.
 - Administered in cases of potential blood mixing (e.g., trauma, ectopic pregnancy).
-

Contraindications

- Allergic to immune globulin.
 - Hemolytic anemia.
 - Anemia.
 - History of coronary artery disease (CAD).
 - Bleeding disorders.
 - High triglycerides.
 - Kidney disease.
 - Diabetes.
-

Nursing Points

1. **Confirm:**
 - Mother is Rh-negative.
 - Baby is Rh-positive.
 - No allergies to immune globulin.
 2. **Informed Consent:** Explain the procedure to the patient and obtain consent if required.
 3. **Administration:**
 - Obtain RhoGam from the blood bank; check lot number and expiration date with another nurse.
 - Administer via IM injection into the ventrogluteal or vastus lateralis muscle.
 4. **Monitoring:**
 - Observe for respiratory difficulties.
 - Check vital signs (v/s).
 - Monitor urine output every 2-4 hours for 8 hours post-administration.
 5. **Side Effects:**
 - Possible fever, chills, back pain, unusual weakness, confusion, slurred speech, or changes in urine color (pink, red, or dark).
 6. **Patient Education:** Provide the mother with an ID card to inform future healthcare providers about her Rh-negative status and the RhoGam administration.
-

Documentation

- Document:
 - Lot number and expiration date.
 - Route of administration.
 - Dose given.
 - Patient education provided.
-

Additional Considerations

- A **positive Coombs test** indicates Rh incompatibility.
- Educate the mother about the risks to the infant if RhoGam is refused (e.g., potential brain damage, hearing and speech issues, seizures, liver and heart failure, shock).